

# PBA: 6. Causality

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# Roadmap

1. What is causality?
2. Randomized experiments
3. Calculating effects
4. Observational studies

Before we begin..  
Rising Interest in 'Causal' Data Science



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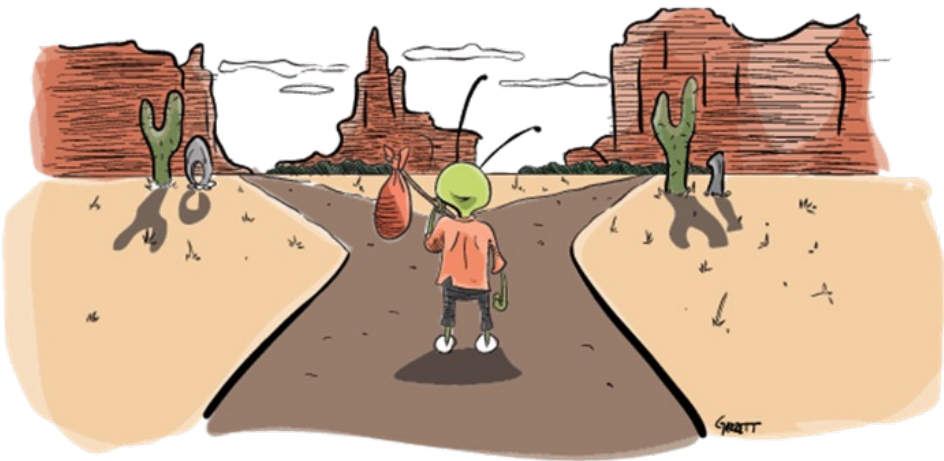
**1/** What is causality?



*The Road Not Taken:*

*"Two roads diverged in a yellow wood,  
And sorry I could not travel both  
And be one traveler, long I stood  
And looked down one as far as I could  
To where it bent in the undergrowth;"*

– Robert Frost



Source: *Chapter 1 of Mastering Metrics* by J. Angrist & J. Pischke

# What is a Causal Effect?

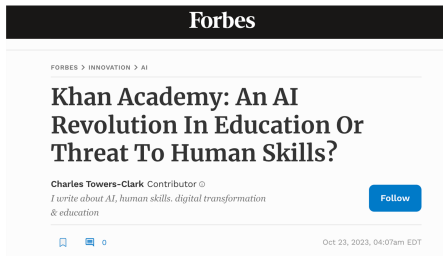
**Factual**

**vs.**

**Counterfactual**

- Does increasing the minimum wage increase the unemployment rate?
  1. Unemployment rate went up after the minimum wage increased.
  2. Would the rate have risen even without the wage increase?
- Does having a CEO with a hobby of flying airplanes lead to better innovation outcomes? ([here](#))
  1. Innovation outcomes are higher for companies Led by Pilot CEOs.
  2. Would the innovation outcomes have been the same w/o a pilot CEO?
- **Fundamental problem of causal inference:**
  - ↪ Can never observe counterfactuals, must be inferred!

# Context: AI Tutoring Study



<https://bit.ly/45VfES8>



Created using DALLE-3

- Can AI assistants improve students' learning outcomes?
- Experimental setting:
  - Randomly assign students to have tutoring sessions with or without AI assistants.
  - AI-assisted sessions focus on adaptive learning techniques.
- Outcome of interest:
  - Improvement in test scores or grades.

# A Tale of Two Students

|           | AI Assistant                  | Test Result |
|-----------|-------------------------------|-------------|
| Student 1 | Tutoring without AI assistant | Fail        |
| Student 2 | Tutoring with AI assistant    | Pass        |

- Q: Did the second respondent pass the standardized exam **because** of the AI assistant?

# Translating into Math

- Useful to have **compact** notation for referring to **treatment variable**:

$$T_i = \begin{cases} 1 & \text{if student } i \text{ had access to tutoring with AI assistant} \\ 0 & \text{if student } i \text{ received tutoring without AI assistant} \end{cases}$$

- Similar notations for the **outcome variable**:

$$Y_i = \begin{cases} 1 & \text{if student } i \text{ passed the standardized test} \\ 0 & \text{if student } i \text{ failed the standardized test} \end{cases}$$

- Note:  $i$  is a placeholder to refer to a generic unit (student in our context):
  - ex)  $Y_{42}$  is the outcome for the 42nd unit.

# A Tale of Two Students (Redux)

|           | AI Assistant                  | Test Result |
|-----------|-------------------------------|-------------|
| Student 1 | Tutoring without AI assistant | Fail        |
| Student 2 | Tutoring with AI assistant    | Pass        |

becomes...

| $i$       | $T_i$ | $Y_i$ |
|-----------|-------|-------|
| Student 1 | 0     | 0     |
| Student 2 | 1     | 1     |

# Causal Effects & Counterfactuals

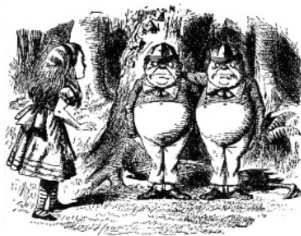
- What does “ $T_i$  causes  $Y_i$ ” mean?  $\rightsquigarrow$  **counterfactuals**, “what if”
- Would the student’s test result change with AI tutoring?
- Two **potential outcomes**:
  - $Y_i(1)$ : would student  $i$  pass the test if s/he received AI-assisted tutoring?
  - $Y_i(0)$ : would student  $i$  pass the test if s/he received tutoring w/o AI?
- **Causal effect**:  $Y_i(1) - Y_i(0)$ 
  - $Y_i(1) - Y_i(0) = 0 \rightsquigarrow$  AI tutoring has no effect on test results
  - $Y_i(1) - Y_i(0) = -1 \rightsquigarrow$  AI tutoring worsens test results
  - $Y_i(1) - Y_i(0) = +1 \rightsquigarrow$  AI tutoring improves test results

# Potential Outcomes

| $i$       | $T_i$ | $Y_i$ | $Y_i(1)$ | $Y_i(0)$ |
|-----------|-------|-------|----------|----------|
| Student 1 | 0     | 0     | ???      | 0        |
| Student 2 | 1     | 1     | 1        | ???      |

- **Fundamental problem of causal inference:**
  - We only observe one of the two potential outcomes.
  - Observe  $Y_i = Y_i(1)$  if  $T_i = 1$  or  $Y_i = Y_i(0)$  if  $T_i = 0$
- To infer causal effect, we need to infer the missing counterfactuals!

# How to Figure Out Counterfactuals?



- Find a similar unit!  $\rightsquigarrow$  **matching**
  - Mill's method of difference
- Did a student pass a standardized test because s/he had access to AI-assisted tutoring?
  - $\rightsquigarrow$  find an identical student who received tutoring w/o AI!
- New Jersey (NJ) increased the minimum wage. Causal effect on unemployment?
  - $\rightsquigarrow$  find a state similar to NJ that didn't increase minimum wage!

# Imperfect Matches

ARNOLD SCHWARZENEGGER DANNY DEVITO  
**TWINS**



Only their mothers can tell them apart...

- The problem: imperfect matches!
- Say we match  $i$  (treated) and  $j$  (control)
- **Selection Bias:**  $Y_i(1) \neq Y_j(1)$
- Those who take treatment may be different than those who take control.
- Q: How can we correct for that?

*Since (1) counterfactuals are never directly observable and (2) perfect matches are rarely attainable, we zoom out and analyze the problem at the aggregate level!*

## **2/** Randomized experiments

# Match Groups, Not Individuals!



- **Randomized control trials:** each unit's treatment assignment is determined by chance.
  - ex) flip a coin or draw red and blue chips from a hat, etc.
- Randomization ensures **balance** between treatment and control group.
  - Treatment and control group are identical **on average**
  - Similar on both observable and unobservable characteristics.

# A Little More Notation

- We will often refer to the **sample size** (number of units) as  $n$ .
- We often have  $n$  measurements of some variable:  $(Y_1, Y_2, \dots, Y_n)$
- Q: How many students in our sample data pass the standardized test?

$$Y_1 + Y_2 + Y_3 + \dots + Y_n$$

- Notation is a bit clunky/heavy, so we often use the **Sigma notation**:

$$\sum_{i=1}^n Y_i = Y_1 + Y_2 + Y_3 + \dots + Y_n$$

- $\sum_{i=1}^n$  means sum each value from  $Y_1$  to  $Y_n$

# Averages

- The **sample average** or **sample mean** is simply the sum of all values divided by the number of values.
- Sigma notation allows us to write this in a compact way:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

- Ex) Suppose we observed 6 students and found out 3 passed the test:

$$\bar{Y} = \frac{1}{6}(1 + 1 + 1 + 0 + 0 + 0) = 0.5$$

# Quantity of interest

- We want to estimate the average causal effects over all units:

$$\begin{aligned}\text{Sample Average Treatment Effect (SATE)} &= \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} \\ &= \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0)\end{aligned}$$

- Q1: Why can't we just calculate this quantity directly?
- What we can estimate instead:

$$\text{Difference in means} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}$$

- $\bar{Y}_{\text{treated}}$ : sample average outcome for treated group
- $\bar{Y}_{\text{control}}$ : sample average outcome for control group
- Q2: When will the difference-in-means be a good estimate of the SATE?

# Why Randomization Works

- Under an RCT, treatment and control groups are random samples.
- Average in the treatment group will be similar to average if all treated:

$$\bar{Y}_{\text{treated}} \approx \frac{1}{n} \sum_{i=1}^n Y_i(1)$$

- Average in the control group will be similar to average if all untreated:

$$\bar{Y}_{\text{control}} \approx \frac{1}{n} \sum_{i=1}^n Y_i(0)$$

- Implies difference-in-means should be close to SATE:

$$\bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}} \approx \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} = \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0) = \text{SATE}$$

# Balance Checking

- Can we determine if randomization “worked”?
- If it did, we shouldn’t see large differences between treatment and control group on **pretreatment variable**.
  - Pretreatment variables are those that are unaffected by treatment (e.g., does receiving AI tutoring affect gender? age?).
- We can check in the actual data for some pretreatment variable  $X$ :
  - $X_{\text{treated}}$ : average value of variable for treated group.
  - $X_{\text{control}}$ : average value of variable for control group.

$\rightsquigarrow$  Under randomization,  $X_{\text{treated}} - X_{\text{control}} \approx 0$

# Multiple Treatments

- Instead of 1 treatment, we might have multiple **treatment arms**:
  - Control condition
  - Treatment A
  - Treatment B
  - Treatment C, etc.
- In this case, we will look at multiple comparisons:
  - $Y_{\text{treated, A}} - Y_{\text{control}}$
  - $Y_{\text{treated, B}} - Y_{\text{control}}$
  - $Y_{\text{treated, A}} - Y_{\text{treated, B}}$
- If treatment arms are randomly assigned, these differences will be good estimators for each causal contrast.

### **3/** Calculating effects

# AI Tutoring Study Data (Simulated)

```
1 # don't forget to load 'dplyr' and 'tidyverse'!  
2 > AI_data <- as_tibble(read.csv(url("https://bit.ly/3FHsusw")))
```

| Variable Name       | Description                                                         |
|---------------------|---------------------------------------------------------------------|
| student_age         | Age of the student in years                                         |
| student_gender      | Gender of the student (1 for Female, 0 for Male)                    |
| treat_ind           | Treatment indicator (1 for treated with AI tutoring, 0 for control) |
| tutor_age           | Age of the tutor in years                                           |
| years_of_experience | Years of tutoring experience                                        |
| education_level     | Tutor's education level                                             |
| test_outcome_pre    | Pre-treatment test outcome (1 for Pass, 0 for Fail)                 |
| test_outcome_post   | Post-treatment test outcome (1 for Pass, 0 for Fail)                |

- This is a ‘simulated’ experimental data we will examine to learn how to calculate the effects using R!
- The simulated data contains information on:
  - Student-level variables (e.g., students’ age, gender)
  - Tutor-level variables (e.g., tutor’s educational level, years of experience)

# Peek at the Data

```
1 > AI_data # Take a peek at the data!
2
3 # A tibble: 500 × 8
4   treat_ind test_outcome_pre test_outcome_post student_age student_gender tutor_age years_of_experience
5   <int>      <int>          <int>      <int>      <int>      <int>      <int>
6 1         1         0            1         11         0         27         17
7 2         1         0            1          6         1         45         10
8 3         0         1            1         15         1         54         14
9 4         0         1            1          6         1         50         18
10 5         0         1            0          8         1         64         13
11 6         1         1            1         15         1         43         20
12 7         1         0            1          7         1         47          6
13 8         1         1            1         16         0         49         14
14 9         0         0            1         12         1         42         11
15 10        1         1            1         11         1         28         11
16 # 490 more rows
17 # 1 more variable: education_level <chr>
18 # Use `print(n = ...)` to see more rows
```

- Again, the question was does AI tutoring help (on average)?

# Compute the Ave. Outcomes in Each Group

- Average outcome for the treated group:

```
1 > treat_mean <- AI_data |>
2   filter(treat_ind == 1) |>
3   summarize(test_outcome_mean = mean(test_outcome_post))
4
5 > treat_mean
6 # A tibble: 1 × 1
7   test_outcome_mean
8             <dbl>
9 1             0.690
```

- Average outcome for the control group:

```
1 > control_mean <- AI_data |>
2   filter(treat_ind == 0) |>
3   summarize(test_outcome_mean = mean(test_outcome_post))
4
5 > control_mean
6 # A tibble: 1 × 1
7   test_outcome_mean
8             <dbl>
9 1             0.651
```

# Calculate the Difference in Means

- Average outcome for the treated group:

```
1 > treat_mean - control_mean
2
3   test_outcome_mean
4 1      0.03903557
```

- Again, this is a good estimate of the causal effect of AI tutoring if and **only if the two groups are statistically identical matches!**
- $\rightsquigarrow$  This would be the case if randomization “worked”.
- We’ll see more ways to do this throughout the semester.

# Checking Balance on Numeric Covariates

- We can use `group_by()` to see how the mean of covariates varies by group:

```
1 > AI_data |>
2   group_by(treat_ind) |>
3   summarize(age_mean = mean(student_age))
4
5 # A tibble: 2 × 2
6   treat_ind age_mean
7   <int>     <dbl>
8     1         11.7
9     2         11.4
```

- Q: How can we do this for all numeric variables in the data? (hint: use `across()` and `where()`)
  - Post your codes on Canvas! (voluntary)

# Checking Balance on Categorical Covariates

- We can group by treatment and categorical control/covariate:

```
1 > AI_data |>
2   group_by(treat_ind, education_level) |>
3   summarize(n = n())
4
5 # A tibble: 9 × 3
6 # Groups:   treat_ind [2]
7   treat_ind education_level      n
8     <int>   <chr>         <int>
9 1         0 Bachelor's degree    52
10 2         0 High school         63
11 3         0 Master's degree     73
12 4         0 Two-year college    73
13 5         1 Bachelor's degree    65
14 6         1 Doctoral/PhD         4
15 7         1 High school         78
16 8         1 Master's degree     18
17 9         1 Two-year college    74
```

⇒ Hard to read!

# pivot\_wider()

- `pivot_wider()` takes data from a single column and moves it into multiple columns based on a grouping variable:

```
1 > AI_data |>
2   group_by(treat_ind, education_level) |>
3   summarize(n = n()) |>
4   pivot_wider(
5     names_from = treat_ind,
6     values_from = n
7   )
8
9 # A tibble: 5 × 3
10   education_level   `0`   `1`
11   <chr>           <int> <int>
12 1 Bachelor's degree    52    65
13 2 High school         63    78
14 3 Master's degree     73    18
15 4 Two-year college     73    74
16 5 Doctoral/PhD        NA     4
```

- `names_from` tells us what variable will map onto the columns.
- `values_from` tell us what values should be mapped into those columns.

# Calculating Diff-in-Means by Group

- The mean differences could be more pronounced for some (sub)group of students compared to others, e.g.,:

```
1 > AI_data |>
2   mutate(
3     treat_ind = if_else(treat_ind == 1, "Treated", "Control"),
4     male_stu = if_else(student_gender == 1, "Male", "Female")
5   ) |>
6   group_by(treat_ind, male_stu) |>
7   summarize(test_outcome_mean = mean(test_outcome_post)) |>
8   pivot_wider(
9     names_from = treat_ind,
10    values_from = test_outcome_mean
11  ) |>
12  mutate(
13    diff_in_means = Treated - Control
14  )
15
16 # A tibble: 2 × 4
17   male_stu Control Treated diff_in_means
18   <chr>      <dbl>   <dbl>         <dbl>
19 1 Female    0.702    0.75      0.0475
20 2 Male      0.607    0.630      0.0231
```

- Q: Anything interesting that stands out to you?

# Some Potential Problems with RCTs

- **Placebo effects:**

- Students experience a benefit from the tutoring solely because they believe AI tutoring is superior, even if the AI isn't actually adding any benefit.
- Q: How could we address this issue?

- **Hawthorne effects:**

- Students change their behavior (e.g., paying more attention, spending more time studying) because they know they are under study.

- Also, **practicality:**

- Can we run experiment (and analyze the data) every time we want to study the effect of some managerial intervention (e.g., new type of advertising)?

## 4/ Observational studies

# Context: Mobile Payment Study

- Data scientists observe ‘naturally assigned’ treatments (e.g., customers’ choice to adopt Line Pay).
  - As we discussed: often can’t randomize for ethical/logistical reasons.



- Can mobile payment services (e.g., Line Pay) change users’ shopping behavior on mobile commerce?
- Q: Why not just compare Line Pay adopters with non-adopters?
  - Post your thoughts on **Canvas!**

# Line Pay Adoption Data (Simulated)

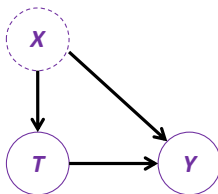
```
1 # don't forget to load 'dplyr' and 'tidyverse'!  
2 > LinePay_data <- as_tibble(read.csv(url("https://bit.ly/3QHEPmM")))
```

| Variable Name  | Variable Description                                                                      |
|----------------|-------------------------------------------------------------------------------------------|
| line_pay_adopt | Indicator for Line Pay adoption (1 for Adopted, 0 for Not Adopted)                        |
| tech_savviness | Binary indicator for technological savviness (0 for Not Tech-Savvy, 1 for Tech-Savvy)     |
| age            | Customer's age, ranging from 18 to 65 years                                               |
| gender         | Customer's gender (0 for Male, 1 for Female)                                              |
| occupation     | Type of occupation (e.g., Student, Engineer, Doctor, Teacher, etc.)                       |
| spending_pre   | Monthly M-commerce spending amount in USD before the launch of Line Pay in the early 2020 |
| spending_post  | Monthly M-commerce spending amount in USD after the launch of Line Pay in the early 2020  |

# Peak at the Data

```
1 > LinePay_data
2
3 # A tibble: 500 × 7
4   line_pay_adopt tech_savviness age gender occupation_type spending_pre spending_post
5             <int>          <int> <int> <int> <chr>          <dbl>      <dbl>
6 1             0            0    23     1 Doctor          58.3      58.3
7 2             1            0    23     0 Doctor          41.3      61.3
8 3             0            0    20     0 Other           43.4      43.4
9 4             0            0    24     1 Other           47.0      47.0
10 5             0            0    25     1 Student          36.5      36.5
11 6             1            0    59     0 Teacher          41.8      61.8
12 7             0            1    32     1 Student          55.2      70.2
13 8             0            0    64     1 Other           58.7      58.7
14 9             0            0    46     0 Engineer         52.6      52.6
15 10            1            1    50     0 Other           61.9      96.9
16 # 490 more rows
17 # Use `print(n = ...)` to see more rows
```

# Confounding



- **Confounder:** pretreatment variable affecting both the treatment & the outcome.
  - Tech-savvy customers (X) are more likely to adopt Line Pay (T).
  - Tech-savvy customers (X) are also more likely to purchase from m-commerce (Y).
- **Confounding bias** in the estimated SATE due to these differences
  - $\bar{Y}_{\text{control}}$  not a good proxy for  $\frac{1}{n} \sum_{i=1}^n Y_i(0)$  in treated group.

# Research Designs

- Q: How Do We Address This Issue?
- Q. restated: How can we find a good comparison group?
  - Depends on the data we have available.
- Three general ways to analyze observational data:
  1. **Cross-sectional design:** compare outcomes of treated and control units at one point in time.
  2. **Before-and-after design:** compare outcomes before and after a unit has been treated, but need over-time data on treated group.
  3. **Difference-in-differences design:** use before/after information for the treated and control group; need over-time on treated & control group.

# Cross-Sectional Design

- Compare Line Pay adopters (treated) and non-adopters (control) after the launch of Line Pay in the early 2020.
  - Customers who adopted Line Pay vs. those who did not.
- Adopters and non-adopters assumed identical on average as in RCT.
- Cross-section comparison estimate:

$$\overline{Y}_{\text{after}}^{\text{treated}} - \overline{Y}_{\text{after}}^{\text{control}}$$

- Could there be confounders? (e.g., Tech-Savvyness)

# Cross-Sectional Design in R

- Sample average outcome for treated & control after adoption:

```
1 > adopted <- LinePay_data |>
2   filter(line_pay_adopt == 1) |>
3   summarize(mean(spending_post)); adopted
4
5 # A tibble: 1 × 1
6   `mean(spending_post)`
7   <dbl>
8 1             81.2
9
10 > no_change <- LinePay_data |>
11   filter(line_pay_adopt == 0) |>
12   summarize(mean(spending_post)); no_change
13
14 # A tibble: 1 × 1
15   `mean(spending_post)`
16   <dbl>
17 1             63.0
```

- Take the difference!:

```
1 > adopted - no_change
2
3   mean(spending_post)
4 1          18.15111
```

# Statistical Control

- **Statistical control:** adjust for confounders using statistical procedures.
  - Can help to reduce confounding bias.
- One type of statistical control: **subclassification**
  - Compare treated and control groups within levels of a confounder.
    - e.g., compare outcomes for a subgroup in which everyone is tech-savvy vs. everyone is NOT tech-savvy.
  - Remaining effect can't be due to the confounder.

# Statistical Control in R

```
1 > LinePay_data |>
2   group_by(tech_savviness, line_pay_adopt) |>
3   summarize(avg_spending = mean(spending_post)) |>
4   mutate(line_pay_adopt = if_else(line_pay_adopt == 1, "adopted", "unadopted")) |>
5   pivot_wider(
6     names_from = line_pay_adopt,
7     values_from = avg_spending
8   ) |>
9   mutate(diff_by_tech_savviness = `adopted` - `unadopted`)
10
11 # A tibble: 2 × 4
12 # Groups:   tech_savviness [2]
13   tech_savviness unadopted adopted diff_by_tech_savviness
14   <int>         <dbl>    <dbl>                <dbl>
15 1         0      50.2     69.7                 19.5
16 2         1      74.6     96.2                 21.6
```

- Limitation: we can only control for observed variables  $\rightsquigarrow$  threat from **‘unmeasured’ confounding**

# Before-and-After Comparison

- Compare m-commerce spending of Line Pay adopters before & after the launch of Line Pay in the early 2020.
- Advantage: all user-specific features held fixed
  - comparing within a user over time.
- Before-and-after estimate:

$$\overline{Y}_{\text{adopters}}^{\text{after}} - \overline{Y}_{\text{adopters}}^{\text{before}}$$

# Before and After Design in R

```
1 > LinePay_data |>
2   filter(line_pay_adopt == 1) |>
3   mutate(
4     spending_change = spending_post - spending_pre
5   ) |>
6   summarize(avg_change = mean(spending_change))
7
8 # A tibble: 1 × 1
9   avg_change
10   <dbl>
11 1      26.5
```

- Limitation: **time-varying confounders**
  - If shopping paradigm was simply moving to mobile commerce (with more people shopping on their phones).

# Differences in Differences

- Key idea: use the before-and-after difference of **control group** to infer what would have happened to **treated group** without treatment.
- DiD estimate:

$$\underbrace{(\bar{Y}_{\text{treated}}^{\text{after}} - \bar{Y}_{\text{treated}}^{\text{before}})}_{\text{trend in treated group}} - \underbrace{(\bar{Y}_{\text{control}}^{\text{after}} - \bar{Y}_{\text{control}}^{\text{before}})}_{\text{trend in control group}}$$

- Change in treated group above and beyond the change in control group.
- **Parallel time trend assumption**
  - Changes in m-commerce spending for Line Pay adopters would be roughly the same as non-adopters if Line Pay wasn't adopted.
  - Limitations: non-parallel trends—what if adopters and non-adopters had different time trends?

# Difference-in-Differences in R

```
1 > LinePay_data |>
2   mutate(
3     spending_change = spending_post - spending_pre,
4     adopted = if_else(line_pay_adopt == 1, "adopted", "unadopted")
5   ) |>
6   group_by(adopted) |>
7   summarize(avg_change = mean(spending_change)) |>
8   pivot_wider(
9     names_from = adopted,
10    values_from = avg_change
11  ) |>
12  mutate(DID = adopted - unadopted)
13
14 # A tibble: 1 × 3
15   adopted unadopted   DID
16   <dbl>    <dbl> <dbl>
17 1    26.5     7.87  18.6
```

# Summarizing Approaches

## 1. **Cross-sectional comparison**

- Compare treated units with control units after treatment
- Assumption: treated and controls units are comparable
- Possible confounding

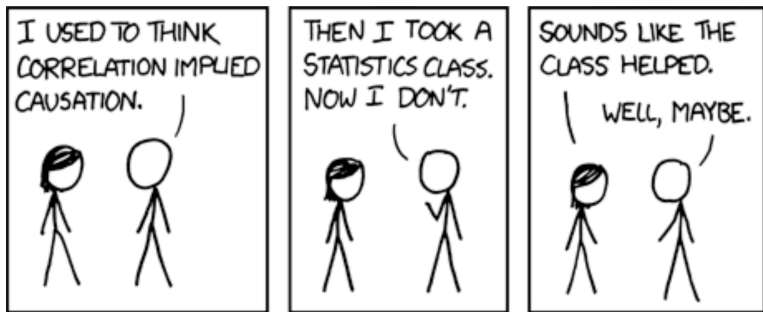
## 2. **Before-and-after comparison**

- Compare the same units before and after treatment
- Assumption: no time-varying confounding

## 3. **Differences-in-differences**

- Assumption: parallel trends assumptions
  - Under this assumption, it accounts for unit-specific and time-varying confounding.
- 
- All rely on assumptions that aren't easily verified.
  - RCTs (experiments) handle confounding by design.

# Causality Understanding Check



**Enjoy your Weekends! :)**

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